

How drops start sliding over solid surfaces

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It has been known for more than 200 years that the maximum static friction force between two solid surfaces is usually greater than the kinetic friction force—the force that is required to maintain the relative motion of the surfaces once the static force has been overcome. But the forces that impede the lateral motion of a drop of liquid on a solid surface are not as well characterized, and there is a lack of understanding about liquid–solid friction in general. Here, we report that the lateral adhesion force between a liquid drop and a solid can also be divided into a static and a kinetic regime. This striking analogy with solid–solid friction is a generic phenomenon that holds for liquids of different polarities and surface tensions on smooth, rough and structured surfaces.

When two solid objects are brought into contact, a threshold force F_{THRD} must be overcome in order for one of the objects to slide^{1–3}. This phenomenon can be visualized in a typical classroom experiment where a solid block attached to a spring is pulled over a solid surface (Fig. 1a). The static force F_s is applied to the stationary block and then increased until it exceeds F_{THRD} , upon which the block begins to slide. After that, a lower kinetic force F_{KIN} is required to maintain the block's motion³. However, it is not clear whether these forces develop in a comparable manner when a drop of liquid resting on a solid surface starts to slide. This gap in our understanding is astonishing, given the fact that liquid drops are omnipresent in our lives and their motion is relevant for numerous applications, including microfluidics⁴, printing⁵, condensation^{6,7}, and water collection^{8,9}. Hence insight on the behaviour of drops that start sliding over solid surfaces is needed.

A sessile drop of liquid is usually in molecular contact with the supporting solid surface. In contrast, two solid bodies are in direct contact only at asperities owing to surface roughness^{10,11}. Thus, the real contact area of a solid–solid contact is much smaller than the apparent contact area. Consequently, the sliding of drops might be fundamentally different. However, by simply observing a drop of water on a pivot window pane, we know that also sessile drops start sliding when a critical tilt angle is reached—that is, when the gravitational force acting on the drop overcomes the lateral adhesion force¹². The question may therefore be raised whether a static and a kinetic regime are also present for sessile drops. The general question is: How do drops start sliding over solid surfaces and how do the forces develop while the drops slide?

Owing to higher gravitational forces, larger drops start sliding at lower tilt angles. Sliding is opposed by capillary forces. They are associated with a contact angle difference between the rear and the front of the drop. Indeed, the interactions between solid surfaces and liquids are described by the liquid–air surface tension γ and the apparent rear and front contact angles of the drop, θ_{Rear} and θ_{Front} , respectively. Thus, the surface tension, the contact angles and the drop contact width L determine the lateral adhesion force F_{LA} by^{13–17}

$$F_{\text{LA}} = kL\gamma (\cos\theta_{\text{Rear}} - \cos\theta_{\text{Front}}) \quad (1)$$

The dimensionless factor k accounts for the precise shape of the solid–liquid–air three-phase contact line of the drop. Values for k were calculated to be between 1/2 and $\pi/2$ (refs 13,18–20).

Despite the omnipresence of drops, the onset of motion has never been correlated with the development of lateral adhesion forces. The lateral adhesion force has been related to external forces that cause a drop to slide, such as gravitational^{21,22}, centrifugal²³, magnetic²⁴, or capillary forces^{25–27}. The contact angles have also been experimentally and numerically investigated for the pinned state—that is, just before and during steady motions^{28,29}. However, once a drop has started to slide, the lateral adhesion force cannot be tracked using simple techniques. Astonishingly, it is unclear how the force develops and how it depends on sliding velocity. We will demonstrate that, for lateral liquid–solid adhesion, we can distinguish a static and a kinetic regime, analogous to solid–solid friction.

To measure the lateral adhesion force between a drop of liquid and a solid substrate, a capillary is positioned in the centre of the drop. The substrate with the drop is moved sideways against the capillary at a constant velocity. When the capillary reaches the edge of the drop, it sticks to the drop. Consequently the motion of the substrate is accompanied by a deformation of the drop as well as a deflection of the capillary. Initially, the drop remains pinned to the substrate (Fig. 1b). Once the capillary exerts a certain critical force, the drop overcomes the lateral adhesion and is set into translational motion relative to the substrate—that is, the front and rear side of the drop start moving. The deflection D of the capillary is measured by recording the position of a reflected laser beam with a position-sensitive detector²⁶. Then, the lateral adhesion force acting on the drop can be calculated by $F_{\text{LA}} = \kappa D$, where κ is the spring constant of the capillary. Simultaneously, the drop's shape is monitored by two cameras, which are synchronized with the force measurement. In this way the lateral adhesion force can be correlated with the contact angles. Velocities ranged from $1 \mu\text{m s}^{-1}$ to 50 mm s^{-1} (see Methods).

As a representative example we start with a drop of an ionic liquid placed on a fluorinated Si wafer. It forms a contact angle of approximately 70° . Moving the wafer laterally increased the force (blue circles in Fig. 2a) until a maximum force of $50 \mu\text{N}$ was reached (after 11 s). This maximum force corresponded to the threshold force for the drop, upon which it started to slide. Then the force

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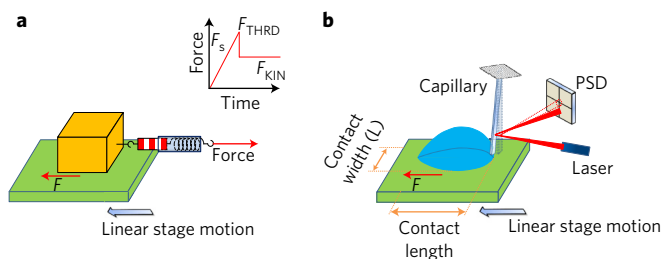


Figure 1 | Schematics of friction force measurements. **a**, Textbook configuration for demonstrating solid–solid friction. **b**, Homemade set-up for measuring liquid–solid friction. A drop of liquid is placed on a solid substrate mounted on a linear stage driven by a step motor. A laser beam incident on the capillary is reflected to a position-sensitive detector (PSD). The contact width between the drop of liquid and the solid surface (orthogonal to the direction of motion) and contact length (parallel to the direction of motion) are simultaneously monitored by cameras (not shown).

subsequently decreased to a constant level of approximately $33 \mu\text{N}$, which was only 66% of the maximal force (after 17 s). Accordingly, we distinguish a static, a transition, and a kinetic regime shown in different shades of green in Fig. 2a. The presence of a higher threshold force compared to the force required for continuous motion explains the manner a drop slides downhill in a tilted-plate experiment. It is analogous to solid–solid friction, where a solid object placed on a tilted plane keeps sliding once the static friction threshold force has been overcome³⁰. Initially, the contact line of the drop was pinned and the contact width and length remained constant at about 2.3 mm (Fig. 2b, first 5 s). Meanwhile, the front and rear contact angles increased and decreased, respectively, as the drop deformed (Fig. 2c). After 5–8 s, the contact line started reforming, resulting in a longer and narrower drop shape. After about 10 s, the contact angles reached a maximum value of approximately 86° at the front and a minimum of approximately 43° at the rear. At this point of maximum contact angle difference of approximately 40° , the static adhesion force reached its maximum, which constituted the threshold force of the translational motion. In the transition regime, the contact length decreased continuously. The rear contact angle increased by more than 10° , whereas the front contact angle decreased by $\approx 5^\circ$ over time. Finally a kinetic regime was reached (after ≈ 18 s), which was characterized by constant lateral adhesion forces, constant contact angles, constant contact widths and contact lengths.

We then took the measured contact widths (Fig. 2b) and contact angles (Fig. 2c) to calculate the lateral adhesion force using equation (1). Thereby we simplified equation (1) by assuming a constant value of $k = 1$, thus disregarding shape changes of the drop contour and variations of the contact angle along the contour¹⁸ (see Supplementary Methods for a discussion of the k factor and its temporal development). This calculation almost quantitatively agreed with the measured lateral adhesion force (red squares in Fig. 2a). In particular, the calculated lateral adhesion force reproduced the transition between the static and kinetic regimes. Once the threshold adhesion force F_{THRD} has been overcome, the drop transitions from a static regime to a steady kinetic regime. Consequently, the liquid–solid adhesion forces in the static and kinetic regimes are substantially different, owing to different contact angles and contact widths in these two regimes:

$$F_s = \gamma L_s (\cos \theta_s^{\text{Rear}} - \cos \theta_s^{\text{Front}}) \quad (2)$$

$$F_{\text{KIN}} = \gamma L_{\text{KIN}} (\cos \theta_{\text{KIN}}^{\text{Rear}} - \cos \theta_{\text{KIN}}^{\text{Front}}) \quad (3)$$

Here, F_s , L_s , θ_s^{Front} and θ_s^{Rear} are the lateral adhesion force, contact width, and front and rear contact angles in the static (S) regime. F_{KIN} ,

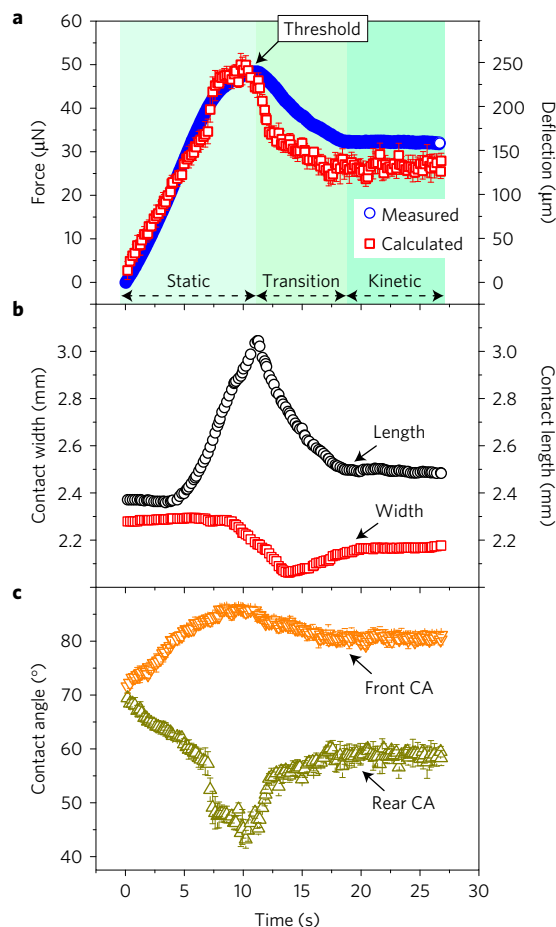


Figure 2 | Lateral adhesion force experiment of a drop of ionic liquid (volume $\approx 1.5 \mu\text{l}$) on a fluorinated silicon wafer. The measurement was performed at a constant linear stage velocity of $200 \mu\text{m s}^{-1}$. The surface tension of the ionic liquid 1-butyl-2,3-dimethylimidazolium bis(trifluoromethanesulfonyl)imide is 34.6 mN m^{-1} . **a**, The measured lateral adhesion force is plotted with blue circles. The error in force measurement is $\pm 1 \mu\text{N}$ and smaller than the symbol size. The calculated force is plotted with red squares. **b**, Simultaneously determined contact width and length of the drop during the force measurement shown in **a**. **c**, Front and rear contact angles during the force measurement. Supplementary Movie 1 shows the motion of the ionic liquid drop on the fluorinated Si wafer during the lateral adhesion force measurement.

L_{K} , $\theta_{\text{KIN}}^{\text{Front}}$ and $\theta_{\text{KIN}}^{\text{Rear}}$ are the lateral adhesion force, contact width, and front and rear contact angles in the kinetic (KIN) regime.

To verify whether the discrimination between static and kinetic lateral adhesion forces is a generic feature, we analysed a number of different liquid–solid combinations (Fig. 3). We used water and hexadecane in addition to a non-volatile ionic liquid (1-butyl-2,3-dimethylimidazolium bis(trifluoromethanesulfonyl)imide). The solid surfaces include superhydrophobic fluorinated silicone nanofilaments (SNFs), fluorinated silicon wafers (Si), fluorinated SU-8 micropillar arrays, fluorinated multilayers of titanium dioxide nanoparticles (TiO_2) and crosslinked polydimethylsiloxane (PDMS) substrates (Supplementary Fig. 1). All samples were homogeneous at the scale of the drop radius and did not have distinct pinning centres. For all liquid–solid combinations we observed a static, a transition and a kinetic regime (Fig. 3a and Supplementary Figs 5–10). It should be noted that the lateral adhesion force in the kinetic regime never exceeded the threshold force (Fig. 3a):

$$F_{\text{THRD}} \geq F_{\text{KIN}} \quad (4)$$

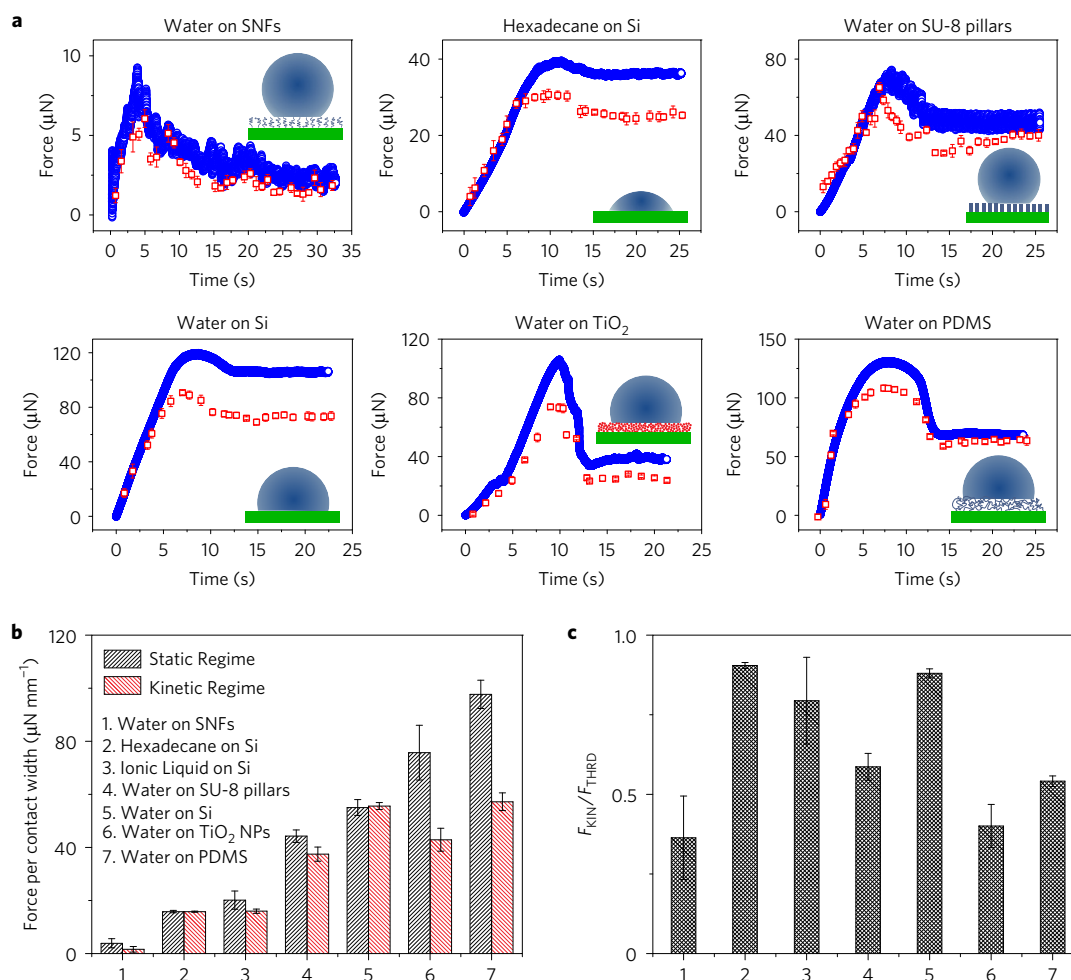


Figure 3 | Lateral adhesion forces for drops of different liquids on solid surfaces. **a**, Measured (blue circles) and calculated lateral adhesion forces using $k=1$ (red squares). The surfaces include silicone nanofilaments (SNFs), silicon wafers (Si), SU-8 square pillar arrays (height, 25 μm; width, 50 μm; centre-centre distance, 100 μm), multilayers of 20-nm titanium dioxide nanoparticles (TiO₂ NPs), and crosslinked polydimethylsiloxane (PDMS). All surfaces except for those of polydimethylsiloxane were fluorinated before use. All fabrication details are provided in the Methods section. The adopted surface tensions of water and hexadecane are 72.5 mN m⁻¹ and 27.5 mN m⁻¹, respectively. The front and rear contact angles in the kinetic regimes were $171^\circ \pm 1^\circ$ and $164^\circ \pm 1^\circ$ for water on fluorinated silicone nanofilaments, $83^\circ \pm 1^\circ$ and $59^\circ \pm 1^\circ$ for hexadecane on fluorinated silicon wafers, $169^\circ \pm 2^\circ$ and $122^\circ \pm 2^\circ$ for water on fluorinated SU-8 pillars, $128^\circ \pm 1^\circ$ and $95^\circ \pm 1^\circ$ for water on fluorinated silicon wafers, $164^\circ \pm 1^\circ$ and $128^\circ \pm 2^\circ$ for water on titanium dioxide nanoparticles, and $121^\circ \pm 1^\circ$ and $81^\circ \pm 1^\circ$ for water on crosslinked PDMS, respectively (Supplementary Figs 5–10). Supplementary Movies 2–7 show the motions of drops on the different surfaces during the lateral adhesion force measurements. **b**, Lateral adhesion force per contact width. Drop volumes between 1.5 and 8.0 μl were chosen to avoid rupturing of drops during motion. **c**, Ratios of the kinetic friction force F_{KIN} and its threshold force F_{THRD} of all liquid–solid combinations that are studied. Here we used F_{KIN} and F_{THRD} prior to normalization with the respective contact widths to point out $F_{\text{KIN}}/F_{\text{THRD}} \leq 1$. Error bars in **b,c** indicate variability between different experiments.

Notably, $F_{\text{THRD}} = F_{\text{KIN}}$ is also possible²⁶. Surfaces coated with brushes of liquid-like poly(dimethylsiloxane) (PDMS) showed no difference between F_{THRD} and F_{KIN} . The reason is that these surfaces show almost identical contact angle hysteresis for water at the threshold and in the kinetic regimes (Supplementary Fig. 11).

Furthermore, good agreement between calculated and measured lateral adhesion forces was observed for various wetting scenarios, including hexadecane and ionic liquid on fluorinated silicon wafers, and water on superhydrophobic silicone nanofilaments and fluorinated SU-8 pillars, where drops partially rested on air cushions. The differences in adhesion between the drops and the substrates resulted in different apparent contact angles and contact widths among the liquid–solid combinations (Supplementary Figs 5–10). We attribute the small discrepancies between calculated and measured lateral adhesion forces to microscopic distortions of the three-phase contact lines¹². The latter can be accounted for by the k factor (Supplementary Figs 5–10 and Supplementary

Table 1). Indeed, the k factor of our measurements lies between 0.25 and 4. It is worth emphasizing that the k factor changed continuously while the drop passed from the static to the kinetic regime. These observations indicate that a constant k factor is not sufficient to describe how drops slide over a solid surface. For large-scale heterogeneous surfaces that exhibit distinct pinning centres, F_{KIN} occasionally exceeded F_{THRD} (Supplementary Fig. 12 and Griffiths and colleagues³¹). Thus, the lateral adhesion force provides quantitative information on the homogeneity or cleanliness of a surface.

To compare the different liquid–solid combinations independently from drop size, we calculated the static and lateral adhesion force per contact width (Fig. 3b). This analysis yielded values ranging from 2 to 110 μN mm⁻¹. The increasing lateral adhesion force per contact width depends on the interplay between surface tension and contact angle hysteresis. In addition, we calculated the ratios of the kinetic force divided by the static threshold force.

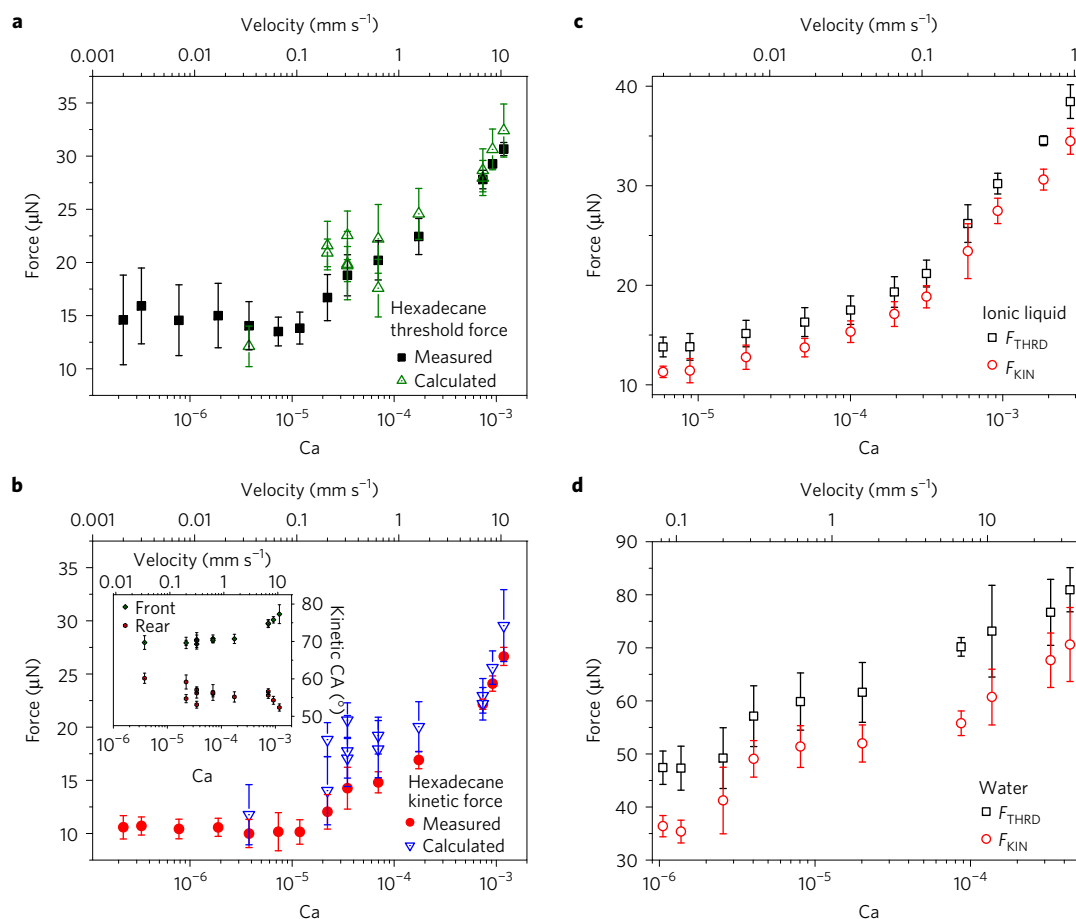


Figure 4 | Velocity dependence of lateral adhesion forces. **a,b**, Development of the lateral adhesion force of hexadecane drops ($\approx 3 \mu\text{l}$) on a fluorinated Si wafer surface at threshold (**a**) and in the kinetic regime (**b**). The inset in **b** shows the front and rear contact angles measured in the kinetic regime. **c,d**, Velocity dependence of lateral adhesion forces of an ionic liquid (1-butyl-2,3-dimethylimidazolium bis(trifluoromethanesulfonyl) imide) drop ($\approx 2 \mu\text{l}$) on a fluorinated Si wafer substrate (**c**) and water drops ($\approx 7.5 \mu\text{l}$) on a fluorinated Si wafer substrate (**d**). Here velocities below 0.08 mm s^{-1} were not studied owing to evaporation of water. The error bars correspond to the standard deviation calculated from five to seven independent measurements. The highest velocity that can be probed is given by the threshold force where the liquid drop detaches from the glass capillary ($\approx 81 \mu\text{N}$ for water, $\approx 36 \mu\text{N}$ for ionic liquid and $\approx 31 \mu\text{N}$ for hexadecane).

For all different liquid–solid combinations, we obtained a ratio of $F_{\text{KIN}}/F_{\text{THRD}} \leq 1$ (Fig. 3c).

In general, contact angles depend on the sliding velocity³². Therefore, we measured F_{KIN} and F_{THRD} in a velocity range from 10^{-3} to 34 mm s^{-1} for hexadecane on a fluorinated silicon wafer surface (filled symbols in Fig. 4a,b). We found that F_{KIN} and F_{THRD} are relatively constant up to a capillary number Ca of 10^{-5} . Then they increase with velocity. Furthermore, the dependence and magnitude of the measured forces are in agreement with the forces calculated by the front and rear contact angles (inset of Fig. 4b) using equation (1) (open symbols in Fig. 4a,b). Similarly, an increase in lateral adhesion forces was observed for ionic liquid and water on fluorinated silicon wafers, respectively (Fig. 4c,d). Perrin *et al.* measured the dynamic macroscopic contact angle of a silicon wafer withdrawn from a silicone oil bath³³. These experiments revealed a strong change in the cosines of the contact angles between $10^{-5} < \text{Ca} < 10^{-3}$. This region marked the transition from energy dissipation dominated by defects to viscous dissipation. The increase in F_{KIN} falls in this regime, and therefore we attribute the increase of F_{KIN} at $\text{Ca} > 10^{-5}$ to viscous dissipation in addition to thermally activated processes at the contact line. The latter was discussed as a result of a rugged energy landscape of the sample surface leading to a thermally activated process of the contact line dynamics³³. In our measurements, we could not resolve a logarithmic dependence of F_{KIN} and F_{THRD} at

$\text{Ca} < 10^{-5}$ which would have given insight into such a thermally activated process of the moving liquid. In particular, we found that the ratio of $F_{\text{KIN}}/F_{\text{THRD}} \leq 1$ holds within the probed velocity range (Supplementary Figs 13 and 14).

To study whether drop–solid friction scales with the contact line or with contact area, we varied the drop volume resulting in a variation of the contact area by a factor of >4 (Supplementary Fig. 13). The linear extrapolation of the kinetic force suggests a linear dependence with an intercept slightly below zero force. Therefore we conclude that the liquid drop friction is dominated by contact line friction and interfacial friction only plays a minor role. This conclusion is supported by an estimation of the friction force contribution of a solid–liquid interface based on a Green–Kubo relation of Huang and Szlufarska³⁴ and by monitoring the motion of drops over the surfaces, showing that the motion is dominated by rolling (Supplementary Fig. 15 and Supplementary Movies 8–11). In contrast, contact line friction does not play a role for liquid–liquid interfaces such as drops on lubricated surfaces³⁵.

As an example of a natural, structured surface, we measured the lateral adhesion of a water drop on a goose feather (Fig. 5a). Again, we observed a static regime and a kinetic regime. The static threshold force was up to three times larger than the average kinetic force required for the continuous motion of the water drop. In addition, the lateral adhesion force varied in the kinetic regime.

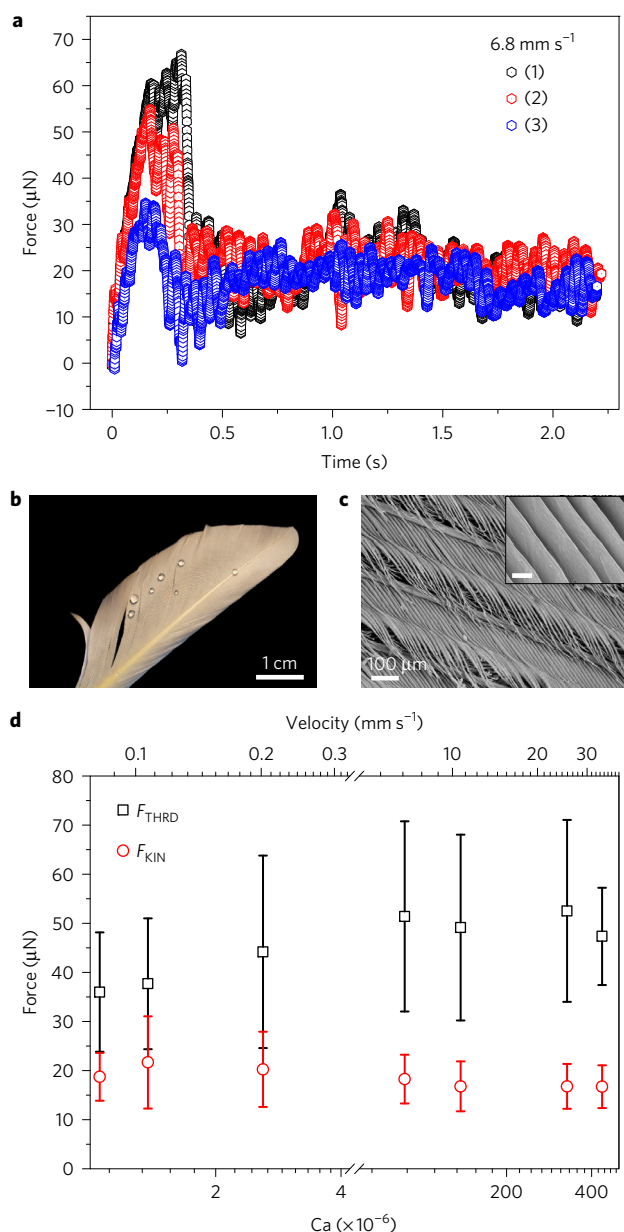


Figure 5 | Lateral adhesion force measurement of a water drop on a goose feather. **a**, Three different lateral adhesion force measurements of a water drop on a goose feather at different areas at a velocity of 6.8 mm s^{-1} (drop volume of $8.0 \mu\text{l}$). The threshold lateral adhesion force varied between $35 \pm 1 \mu\text{N}$ and $67 \pm 2 \mu\text{N}$. Continuing the kinetic motion of the water drop required a force up to $35 \mu\text{N}$. **b**, An optical image of the feather with resting water drops. **c**, Scanning electron microscopy images reveal the detailed structure of the feather. The inset is an enlarged area showing the microscopic structure. The scale bar in the inset is $10 \mu\text{m}$. **d**, No velocity dependence of the lateral adhesion force at threshold (black squares) and in the kinetic regime (red circles) could be measured. The error bars correspond to the standard deviations calculated from five to seven measurements each.

It is likely that these variations are related to the flexibility and the microscopic structures of the feather (Fig. 5b,c), which lead to a stick-slip motion of the drop. Energy is dissipated mainly by depinning from a few surface protrusions which shows no velocity dependence of the lateral adhesion force at the threshold and in the kinetic regime (Fig. 5d and Supplementary Fig. 16)³⁶. The latter is in contrast to stick-slip motion of two solids^{37,38}. Consequently,

for such rough and elastic surfaces the contribution of viscous dissipation to the overall energy dissipation of the sliding drop is low. It is not clear what role such a stick-slip motion and the associated velocity independence play for the life of a goose in water. The constant and low kinetic force could potentially reduce energy losses at the three-phase contact line of the feather with water and air.

Methods

Methods, including statements of data availability and any associated accession codes and references, are available in the [online version of this paper](#).

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Author contributions

N.G. carried out the experiments and wrote the manuscript. D.W.P., N.G., R.B. and H.-J.B. designed and constructed the homebuilt set-up. F.G. and S.W. prepared the solid surfaces. R.B., D.V., N.G. and H.-J.B. contributed to the experimental planning, data analysis, and manuscript preparation. All authors reviewed and approved the manuscript.

Additional information

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Competing financial interests

The authors declare no competing financial interests.

Methods

Preparation of solid surfaces and liquid drops. The following chemicals were used to fabricate the solid substrates: trichloromethylsilane (TCMS, 99%, Sigma-Aldrich), 1H,1H,2H,2H-perfluorodecyltrichlorosilane (PFOTS, 96%, Alfa Aesar), 1H,1H,2H,2H-perfluorooctyltrichlorosilane (PFOTS, 97%, Sigma-Aldrich), Sylgard 184 (PDMS precursor, Dow Corning), Sylgard 184 curing agent (Dow Corning), trimethylsiloxy-terminated poly-(dimethylsiloxane) (PDMS, 200 cSt, ABCR), sulfuric acid (95%, VWR), ammonia solution (28%, VWR), hydrogen peroxide solution (Sigma-Aldrich, 34.5–36.5%), *n*-hexane (99.99%, Fisher Chemical), toluene (99.99%, Fisher Chemical), isopropanol (99.5%, Fisher Chemical), acetone (99.98%, Fisher Chemical), ethanol (absolute, 99.96%, VWR Chemicals), tetrahydrofuran (99.9%, Sigma-Aldrich), SU-8 3025 photoresist (Microchem), mr-Dev 600 developer (micro resist technology), 18NR-T (Dyesol) and sodium hydroxide (98%, Sigma-Aldrich). Reagents were used as received. Polished silicon wafers were obtained from Si-Mat. Thin glass slides of $24 \times 60 \text{ mm}^2$ and $170 \pm 5 \mu\text{m}$ thickness were obtained from Carl-Roth. All syntheses were carried out at room temperature. The liquids drops used for force measurements included those of hexadecane, 1-butyl-2,3-dimethylimidazolium bis(trifluoromethanesulfonyl)imide and water. Hexadecane (99%, Sigma-Aldrich) and 1-butyl-2,3-dimethylimidazolium bis(trifluoromethanesulfonyl)imide (99%, iolitec) were used as received. Water with a typical resistivity of $18.2 \text{ M}\Omega \text{ cm}$ was obtained from a Sartorius Arium 661 VF Water Purification System.

Silicone nanofilaments. Silicone nanofilaments were synthesized by modifying previously reported methods^{39,40}. The fabrication contained the following steps. First, 120 μl of TCMS was added into a reaction chamber containing 50 ml of water-saturated *n*-hexane. The resulting solution was stirred for 60 s. Then, cover glass slides were immersed in the solution, before sealing the reaction chamber. After 3 days the TCMS-coated glass slides were removed from the reaction chamber, rinsed with *n*-hexane and dried under a nitrogen stream. The TCMS-coated glass slides were activated in an oxygen plasma chamber (75 W, 2 min, $7 \text{ cm}^3 \text{ min}^{-1}$ oxygen flow rate, Diener Electronic Femto) before modification with the semi-fluorinated silane PFOTS. Therefore, the activated substrates were immersed in a solution containing 25 μl of PFOTS dispersed in 50 ml of *n*-hexane for 20 min. Subsequently, the substrates were rinsed with *n*-hexane and dried under a nitrogen stream. Representative scanning electron microscopic images of the resulting silicone nanofilaments are shown in Supplementary Fig. 1a,b.

Silicon wafers. Pristine silicon wafers were cleaned by ultrasonication in toluene, acetone and ethanol for 15 min, respectively. Subsequently, the pre-cleaned wafers were plasma-cleaned and activated in an oxygen plasma chamber (300 W, 10 min, $6 \text{ cm}^3 \text{ min}^{-1}$ oxygen flow rate). To fluorinate the surfaces, 25 μl of PFOTS was mixed with 50 ml of *n*-hexane. The activated silicon wafers were immersed in the solution for 5 to 15 min. Afterwards, the silicon wafers were rinsed with *n*-hexane, followed by ultrasonication in isopropanol for 5 min. The cleaned silicon wafers were dried under a nitrogen stream. It should be noted that the immersion time influences the lateral adhesion force of the fluorinated silicon wafers. It should be noted that the immersion time influences the lateral adhesion force of the fluorinated silicon wafers. To access higher sliding velocities we improved the preparation of the Si wafers. We found that ultrasonication in isopropanol helps to remove residues from the fluorination. Generally, shorter fluorination times (5 min) yielded a lower adhesion. Therefore the lateral adhesion forces measured for fluorinated silicon wafers in Fig. 4 are lower compared to Fig. 2. Lower lateral adhesion forces allowed us to investigate higher velocities. The scanning electron microscopy analysis revealed featureless, smooth surfaces, as shown in Supplementary Fig. 1c,d.

SU-8 pillars. SU-8 micropillar arrays were prepared on thin glass slides by photolithography⁴¹. The square pillars were designed $25 \mu\text{m}$ high, with $50 \times 50 \mu\text{m}^2$ top areas. The pillar-pillar distance between centres of two adjacent pillars in a row was $100 \mu\text{m}$. The fabrication process consisted of the following steps. First, glass slides (Carl-Roth) were cleaned by ultrasonication in tetrahydrofuran, acetone and ethanol for 15 min, respectively. Then, SU-8 photoresist was spin-coated (500 r.p.m. for 5 s followed by 3,000 r.p.m. for 30 s, SÜSS MicroTec) on the glass slides. The coated glass slides were heated at 65°C for 3 min, 95°C for 10 min and then 65°C for 30 min, respectively. Subsequently, the samples were slowly cooled down within two hours and exposed to ultraviolet light (mercury lamp, 350 W) under a photolithography mask for 15 s (masker aligner SÜSS MicroTec MJB3 UV400). To crosslink the photoresist, the samples were heated up again at 65°C for 3 min, 95°C for 10 min and 65°C for 30 min, and then cooled down slowly. Next, the samples were immersed in the SU-8 developer solution for 5 min, washed with isopropanol and then dried in air, resulting in surfaces coated with SU-8 pillars. The dried samples were immersed in 0.1 M NaOH solution overnight to hydrolyse the surfaces of the SU-8 pillars. After rinsing the hydrolysed surfaces with water and ethanol, the samples were immersed in a solution containing 25 μl of PFOTS dispersed in 50 ml of *n*-hexane for 20 min.

Finally, the fluorinated SU-8 micropillar surfaces were rinsed with *n*-hexane and dried under a nitrogen stream. Representative scanning electron microscopic images of an obtained SU-8 micropillar array are shown in Supplementary Fig. 1e,f.

PDMS. Crosslinked PDMS surfaces: Sylgard 184 PDMS precursor was mixed with Sylgard 184 curing agent in a ratio of 10:1 by weight and then degassed three times in a vacuum chamber for 5 min, respectively. Afterwards, the viscous mixture was deposited on glass slides and cured at 60°C overnight. The scanning electron microscopy analysis revealed featureless, smooth surfaces, as shown in Supplementary Fig. 1g,h.

So-termed 'liquid-like' PDMS samples were prepared following the method reported by Krumpfer and McCarthy⁴². Pre-cleaned cover slides were placed in a 2:1 mixture of sulfuric acid and hydrogen peroxide solution, placed in a water bath, and held at 70°C for 2 h. Then the samples were rinsed with pure water and ethanol. After being dried, the cover slides were put into 500 ml ultraclean borosilicate glass vials (Duran Pure vials GL45 with a dust cover, Duran Group, Schott) and sufficiently wetted with PDMS. The vials were resealed and placed in an oven at 100°C for 24 h. Then the samples were thoroughly rinsed with *n*-hexane, acetone, and Milli-Q water, and then dried.

The flexibility of the O–Si–O bonds confers high mobility to PDMS chains. As only one end of the PDMS is covalently grafted on the substrate, the remaining part of the PDMS keeps its high mobility with rotational and/or bending motion. Thus, these grafted layers are rotationally dynamic and behave 'liquid-like'. Water drops slide off these 'liquid-like' surfaces when tilting the surface by 4° (drop volume: 20 ml) (ref. 42).

Titanium oxide nanoparticles. TiO_2 surfaces consisting of $\sim 20\text{-nm}$ TiO_2 nanoparticles (NPs) were fabricated from TiO_2 nanoparticle paste (18NR-T, Dyesol)⁴³. First, TiO_2 nanoparticle paste was coated on a silicon wafer by doctor blade coating with $50 \mu\text{m}$ spacers. Then, the coated wafer was sintered at 500°C for 30 min, resulting in nano-roughened TiO_2 structures. After cooling down, the TiO_2 structures were modified with PFOTS by chemical vapour deposition (CVD) in a desiccator for 2 h in vacuum. Scanning electron microscopic images of a resulting TiO_2 surface are shown in Supplementary Fig. 1i,j.

Force measurements. Sensitivity of the laser deflection system. For the measurement of the lateral adhesion force by means of our laser deflection system illustrated in Fig. 1b, a hollow rectangle capillary was used to move a liquid drop against a substrate underneath^{26,44}. A laser beam was incident on the capillary, which reflected the beam to the centre of a position-sensitive detector (PSD). The capillary was manufactured in borosilicate glass (VitroCom's Vitrotubes, CM Scientific). It was 35 mm long. The dimensions of the capillary were $0.04 \text{ mm} \times 0.40 \text{ mm}$, with a wall thickness of 0.028 mm . To increase the reflectivity of the capillary, its wide side was sputter-coated with a 5-nm-thick layer of chromium and a 50-nm-thick layer of gold. The chromium was used as an adhesion layer for the gold. When the laser hit the PSD, four photocurrents were generated by the PSD at two opposite sides along the horizontal axis and two opposite sides along the vertical axis. These four photocurrents were collected and converted into four voltages (U_{x1} , U_{x2} , U_{y1} , and U_{y2}) using an amplifier with a current-to-voltage converter. These four voltages were read out in two values by the amplifier corresponding to the horizontal (x) and vertical (y) coordinates of the laser spot on the PSD:

$$U_x = (5V) \frac{U_{x1} - U_{x2}}{U_{x1} + U_{x2}} \quad (5)$$

$$U_y = (5V) \frac{U_{y1} - U_{y2}}{U_{y1} + U_{y2}} \quad (6)$$

The divisions were performed by an analog divider (Burr-Brown DIV100). $U_y = U_{\text{PSD}}$ and corresponded to the deflection of the capillary in the measurement of the lateral adhesion force. These voltage values were digitalized in an AD converter (National Instruments (NI)). Furthermore, the data acquisition software generated triggers for the cameras to ensure synchronization between the cameras and the PSD signals.

The relationship between the deflection of the capillary, D , and the generated U_{PSD} represents the sensitivity S of the laser deflection system:

$$S = \frac{U_{\text{PSD}}}{D} \quad (7)$$

For calibration a defined deflection of the capillary was applied close to its free end by a micromanipulator. A linear correlation was found between U_{PSD} and D for deflections up to $800 \mu\text{m}$. The corresponding sensitivities were in the range from -4 to $-5 \times 10^{-3} \text{ V } \mu\text{m}^{-1}$. We performed a calibration for each new capillary and after each readjustment of the set-up. Supplementary Fig. 2 shows a representative plot, with $S = -4.16 \times 10^{-3} \text{ V } \mu\text{m}^{-1}$ and Pearson $r = 0.9999$. The sign of the voltage corresponds to the direction of the laser displacement on the PSD along the y -axis.

Spring constant of the capillary. The spring constant κ of the glass capillary was calibrated by measuring the force acting on the capillary with respect to the lateral deflection. First, the capillary was fixed at one end into a copper holder as a fixture (Supplementary Fig. 3). Then the entire fixture was mounted to a micromanipulator (MMO-203, Narishige), which allowed us to move the fixture precisely (with 1 μm precision). A pin with supporting pedestals at the bottom and a pointy tip facing up was placed on a microgram balance (Supplementary Fig. 3). Using the micromanipulator, the free end of the capillary was located right above the tip of the pin. Afterwards, the capillary was lowered down slowly by means of the micromanipulator to engage with the tip until a slight deflection of the capillary was generated. The place where the free end of the capillary was contacted by the pin was within 1 ± 0.2 mm from the edge of the free end.

The microgram balance read out the weight that reflected the force, F , between the pin and the deflected capillary, as $F = mg$, where $g = 9.81 \text{ m s}^{-2}$. Supplementary Fig. 4 is a representative plot of the calibration of the spring constant of the glass capillary. The relation between the force, F , and the deflection, D , followed Hooke's law⁴⁵

$$F = \kappa D \quad (8)$$

with $\kappa = 0.202 \pm 0.002 \text{ N m}^{-1}$.

Force conversion. To obtain the lateral adhesion force, the voltage values directly collected from the PSD need to be converted into forces. Combining equations (7) and (8) yielded the relationship between the lateral adhesion forces, F_{LA} , and the generated voltage:

$$F_{\text{LA}} \equiv F = \kappa D = \kappa \frac{U_{\text{PSD}}}{S} \quad (9)$$

Substrate movement system. Drops were deposited on solid substrates using a pipette (Thermo Scientific, Finn timer F2). To probe a velocity regime that spans five orders of magnitude we implemented three different motorized stages to drive the relative motion between the drop and the substrate. The low-velocity regime of 2 to 200 $\mu\text{m s}^{-1}$ was realized by a slow linear stage consisting of a custom-made step motor whose single steps are approximately 34.2 nm. The medium-velocity regime of 0.2 to 1.6 mm s^{-1} was achieved by using a circular disk driven by a linear motor (Faulhaber 2232 A 012 SR with transmission gear 22 EK 23014:1). The circular disk rotated at a constant angular velocity in the regime of 0.02 to 0.15 r.p.m. The liquid drops were placed at a radius of 100 mm to the rotation centre. The high-velocity regime of 6.8 to 33.9 mm s^{-1} was realized again by a linear stage consisting of a custom-made step motor in which single steps are approximately 5 μm .

Drop imaging. A high-speed camera (FASTCAM Mini UX100 type 800K-M-8G, Photron) equipped with a manual zoom lens was placed orthogonal to the drop motion. Its lens was focused on the drop with the narrow side of the capillary facing this orthogonal camera to monitor the change in front and rear contact

angles as well as contact length. A second camera (Marlin F131B, Allied Vision) equipped with a manual zoom lens was placed parallel to the drop motion. Its lens was focused on the drop with the wide side of the capillary facing the parallel camera to monitor the change in contact width. Using the cameras, contact angles above approximately 160° are hardly accessible due to the narrow gap between the liquid and the solid. Therefore the values do not reflect the real contact angles but should be considered as a lower boundary. On superhydrophobic surfaces the advancing contact angle should always be close to 180° ⁴⁶.

Before the sample was moved, the glass capillary was placed in the centre of the drop. Both cameras were synchronized in time with the data acquisition system. Throughout an experiment the synchronization between force and optical measurements via the cameras was < 60 ms. The drop was illuminated with two cold-light sources (KL 2500 LCD, Zeiss; L2, Leica), each of them located opposite to a camera, respectively. The contact angles, as well as the contact widths and lengths, were measured by means of ImageJ. The contact widths and lengths were calculated using the known dimensions of the capillary as scale. Supplementary Figs 5–10 show representative lateral adhesion forces with measured contact widths, contact lengths, and contact angles corresponding to the cases presented in Fig. 3a. Supplementary Movies 1–7 show the motions of drops on different surfaces during the lateral adhesion force measurements corresponding to the cases presented in Figs 2 and 3.

Data availability. The data that support the plots within this paper and other findings of this study are available from the corresponding author upon request.

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